

Lecture 8b

EE-215 Electronic Devices and Circuits

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Discrete-Circuit MOS Amplifiers

- Thus any of the above biasing methods can be utilized to implement a MOSFET amplifier
 - A MOSFET amplifier can be a
 - Common Source Amplifier
 - Common Gate Amplifier
 - or a Source Follower
 - here we will utilize the constant-current biasing scheme,
 - to implement the three basic configurations of MOS amplifiers

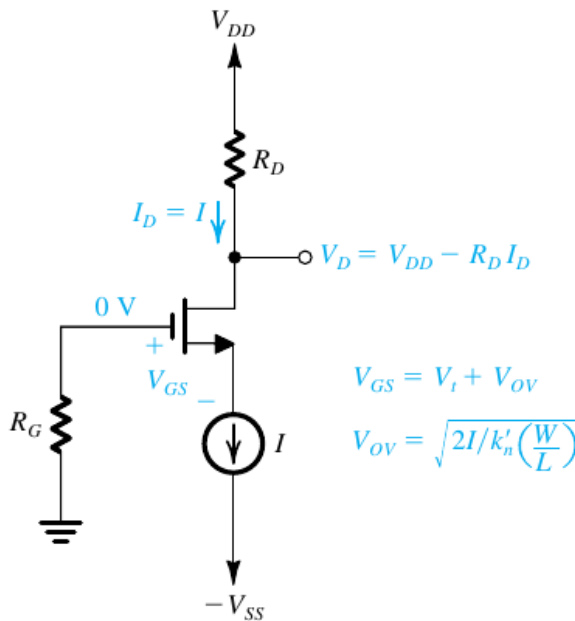
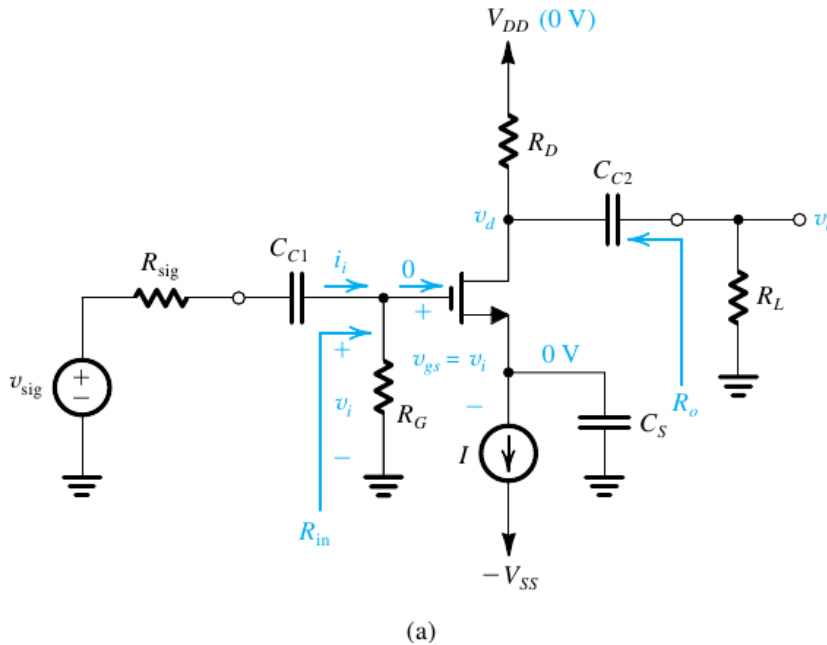


Figure 5.56 Basic structure of the circuit used to realize single-stage, discrete-circuit MOS amplifier configurations.

The Common-Source (CS) Amplifier

- A common-source amplifier realized using the constant-current biasing is shown here

- here a signal ground (or ac ground) at the source is
 - realized by connecting a capacitor C_S between the source and ground
 - the value of C_S should be large enough so as to provide a very low impedance at the desired frequencies (i.e. $\frac{1}{j\omega C_S}$ should be very small)



- this capacitor C_S thus acts as a short circuit for the ac signal frequencies
 - the signal current passes through C_S to ground and thus bypass the output resistance of current source I
 - C_S is therefore called a bypass capacitor
 - C_{C1} is used to couple the ac signal source to the gate of MOSFET
 - this capacitor blocks dc and acts as a short circuit at ac and is called a coupling capacitor
- C_{C2} is also a coupling capacitor which is utilized to connect the load to the drain terminal
 - As a consequence of the coupling and bypass capacitors,
 - the dc equivalent circuit stays the same as the basic constant-current source biasing circuit
- As a consequence of the coupling and bypass capacitors,
 - the dc equivalent circuit stays the same as the basic constant-current source biasing circuit

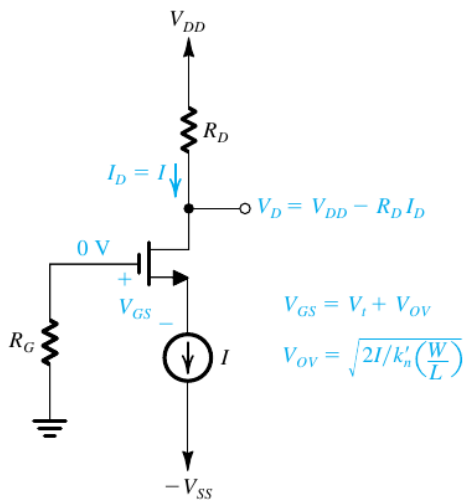


Figure 5.56 Basic structure of the circuit used to realize single-stage, discrete-circuit MOS amplifier configurations.

- Now to determine the terminal characteristics of the CS amplifier (i.e. R_{in} , R_o , A_v , G_v),
 - we replace the CS amplifier by its equivalent small-signal circuit
 - i.e. we replace the MOSFET by its small-signal model
 - suppress the dc sources
 - replace capacitors by short-circuits (if $\frac{1}{j\omega C} \rightarrow 0$)

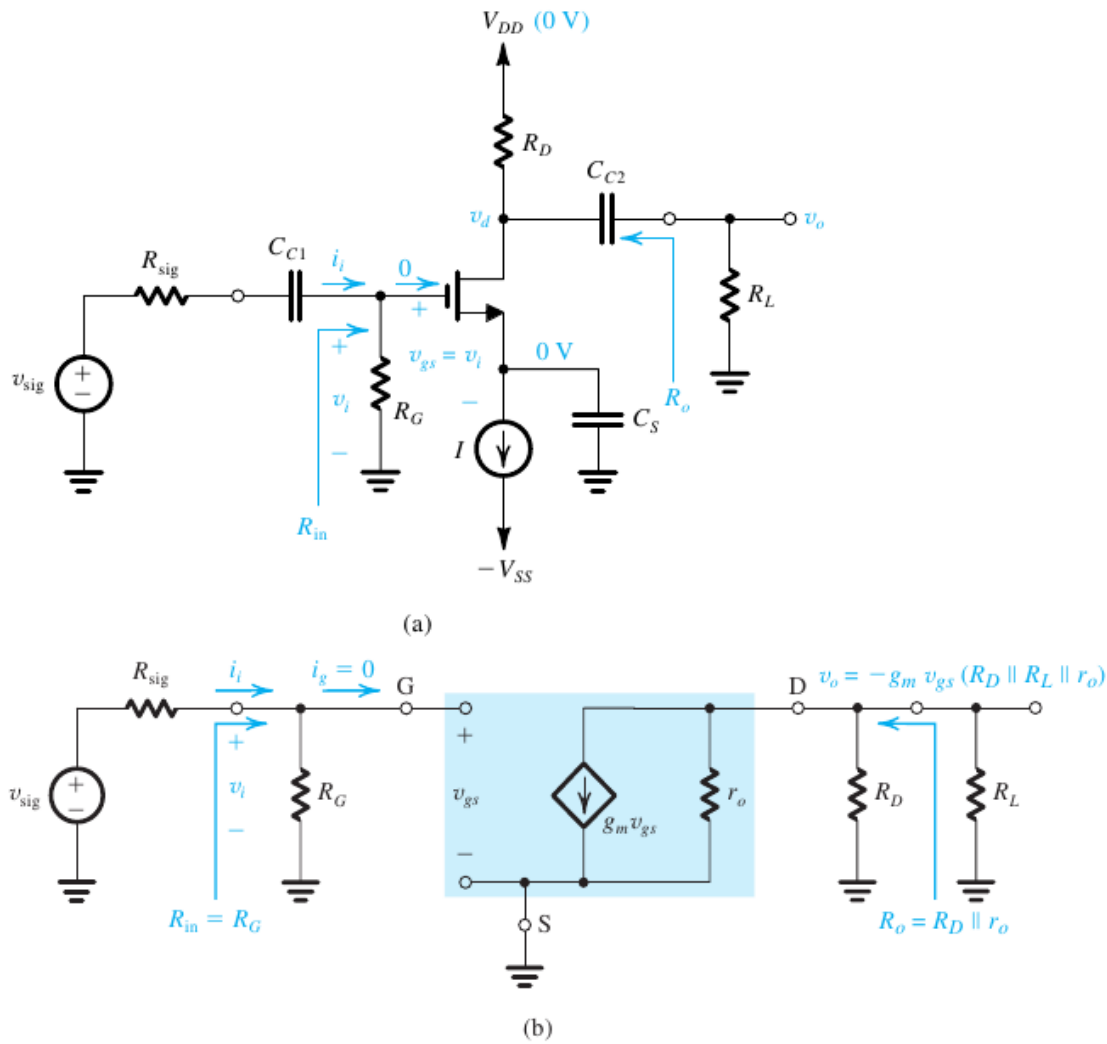
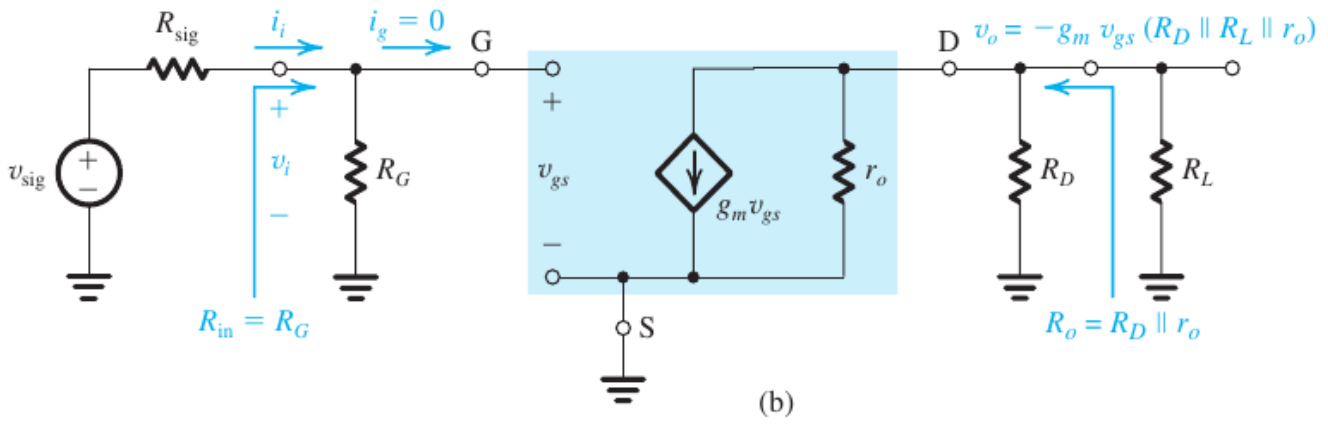


Figure 5.57 (a) Common-source amplifier based on the circuit of Fig. 5.56. (b) Equivalent circuit of the amplifier for small-signal analysis.

- here the input resistance , $R_{in} = R_G$
 - to determine R_o
 - we apply a test source at the output, suppress the independent sources i.e. $v_i = 0$
 - $v_{gs} = v_i$
 - $v_i = 0 \Rightarrow v_{gs} = 0 \Rightarrow g_m v_{gs} = 0$
 - $\Rightarrow R_o = R_D \parallel r_o$



- $A_v = \frac{v_o}{v_i}$

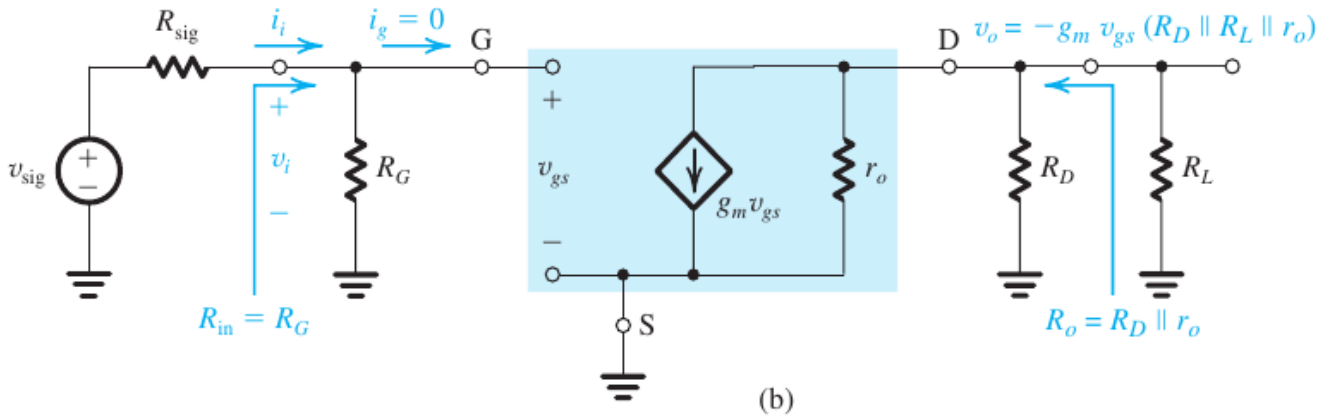
- as r_o , R_D and R_L are in parallel

- $\Rightarrow -g_m v_{gs} = \frac{v_o}{r_o \parallel R_D \parallel R_L}$

- or $-g_m (r_o \parallel R_D \parallel R_L) = \frac{v_o}{v_{gs}}$

- As $v_{gs} = v_i \Rightarrow A_v = \frac{v_o}{v_i} = -g_m (r_o \parallel R_D \parallel R_L)$

- and $A_{vo} = \left. \frac{v_o}{v_i} \right|_{R_L \rightarrow \infty} = -g_m (r_o \parallel R_D)$



- $G_v = \frac{v_o}{v_{sig}} = \frac{v_i}{v_{sig}} \frac{v_o}{v_i} = \frac{v_i}{v_{sig}} A_v$

- by voltage divider $v_i = \frac{R_G}{R_G + R_{sig}} v_{sig}$

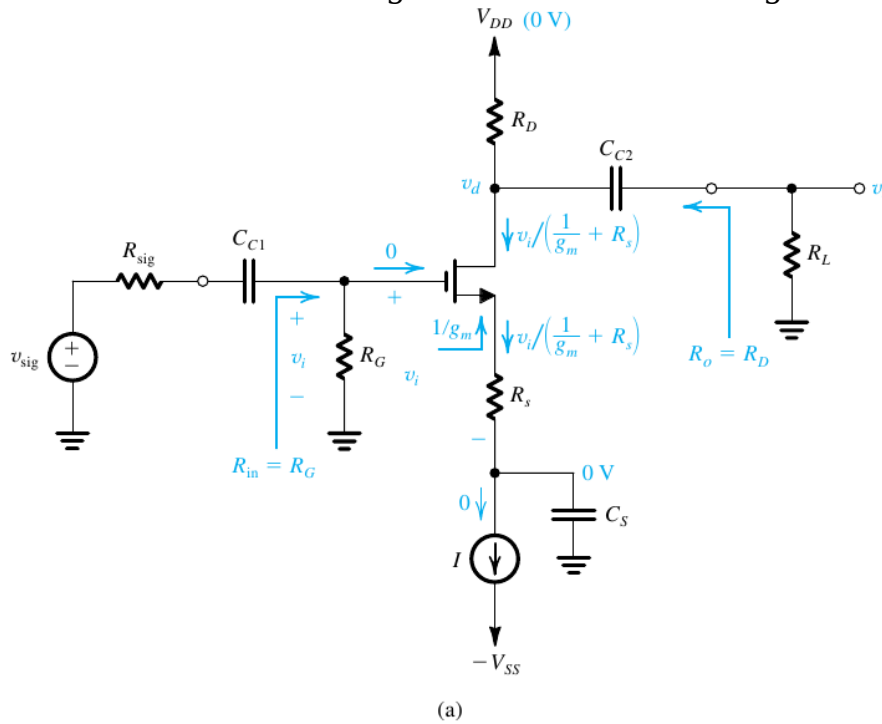
- $\Rightarrow G_v = \frac{v_i}{v_{sig}} A_v = \frac{R_G}{R_G + R_{sig}} A_v$

- As $A_v = -g_m (r_o \parallel R_D \parallel R_L)$

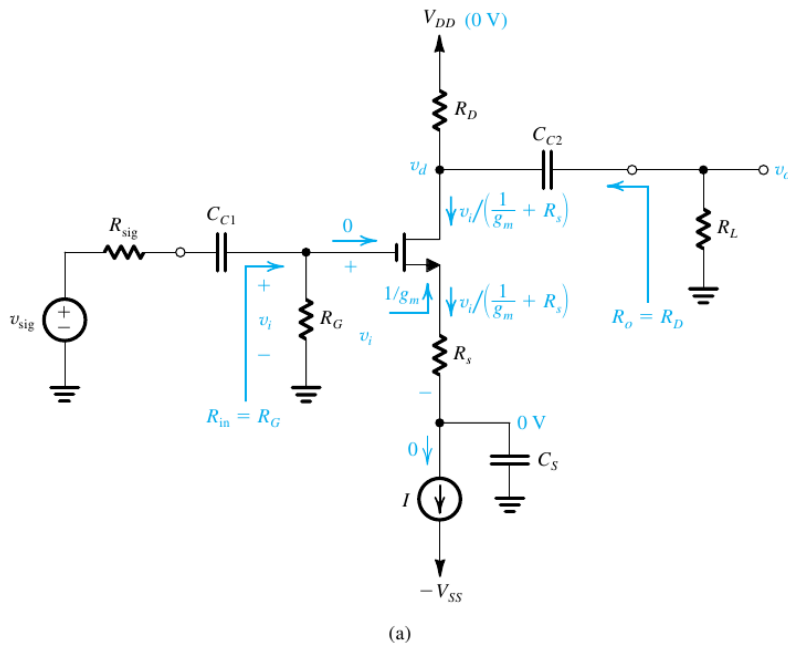
- $\Rightarrow G_v = \left(\frac{R_G}{R_G + R_{sig}} \right) \left[-g_m (r_o \parallel R_D \parallel R_L) \right]$

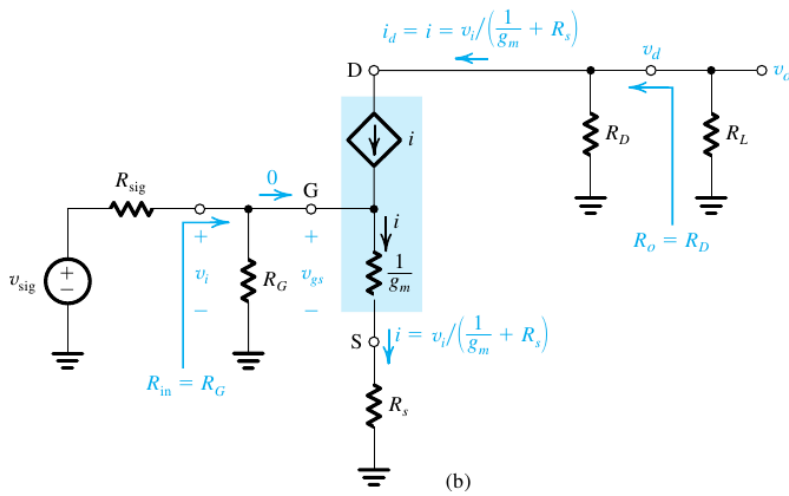
The Common-Source Amplifier with a Source Resistance

- the Common-source amplifier with a source resistance, utilizing the constant-current biasing scheme is shown in figure

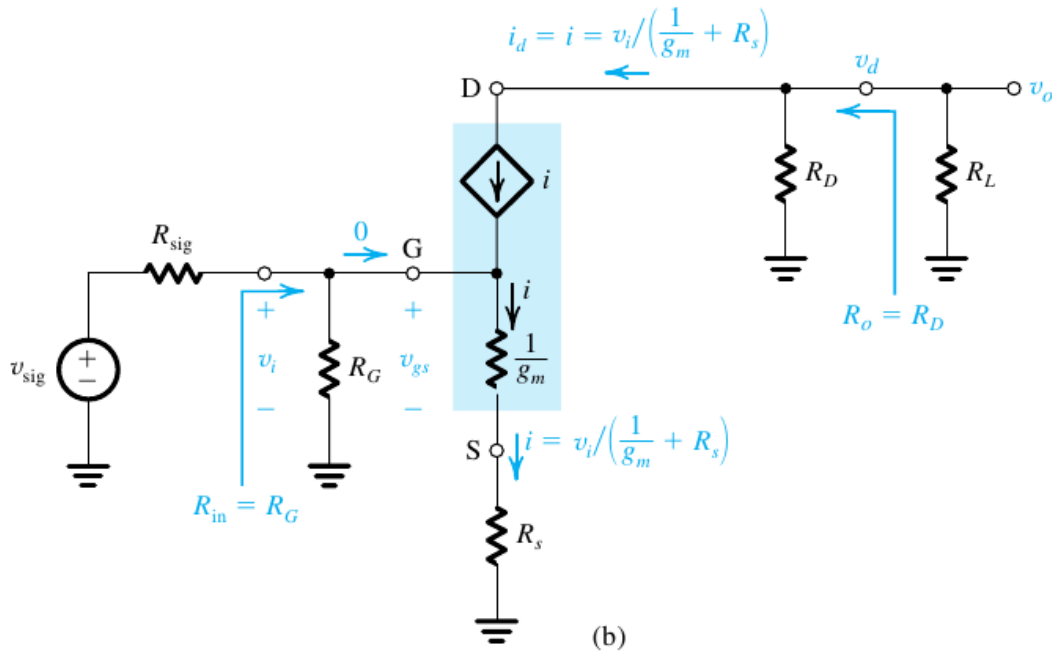


- small signal equivalent circuit for fig a, can be given as





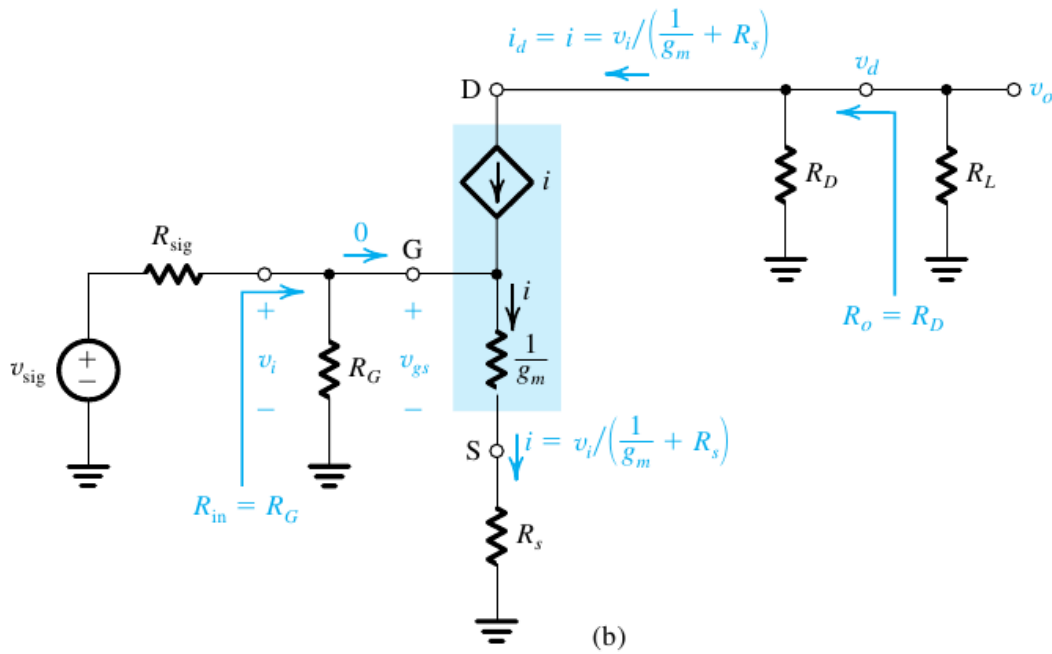
- here $R_{in} = R_G$



- here $v_{gs} = \frac{1}{\frac{1}{g_m} + R_s} v_i$, $v_i = \frac{R_G}{R_G + R_{sig}} v_{sig}$

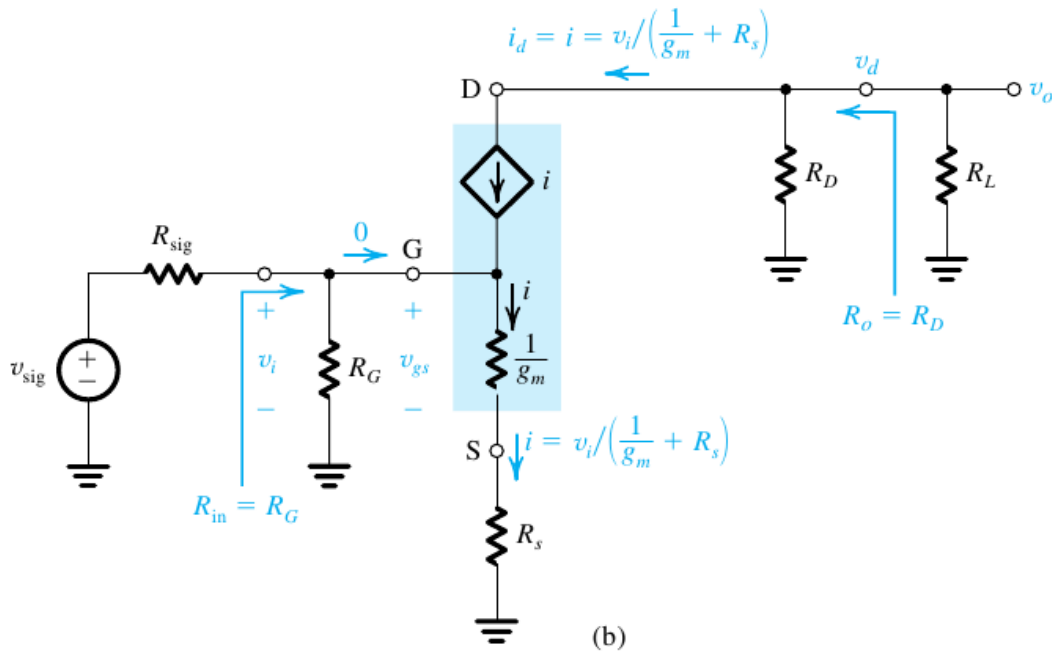
$$\circ \Rightarrow v_{gs} = \frac{1}{\frac{1}{g_m} + R_s} \frac{R_G}{R_G + R_{sig}} v_{sig} \Rightarrow \text{if } v_{sig} = 0 \text{ then } v_{gs} = 0$$

- to determine R_o , we set $v_i = 0 \Rightarrow v_{gs} = 0 \Rightarrow g_m v_{gs} = 0$ or $i = 0$
- $\Rightarrow R_o = R_D$



$$\bullet i = \frac{v_i}{\frac{1}{g_m} + R_s} = -\frac{v_o}{R_D \parallel R_L}$$

$$\circ \Rightarrow \frac{v_o}{v_i} = \frac{-R_D \parallel R_L}{\frac{1}{g_m} + R_s} = A_v \Rightarrow A_{vo} = A_v|_{R_L \rightarrow \infty} = \frac{-R_D}{\frac{1}{g_m} + R_s}$$

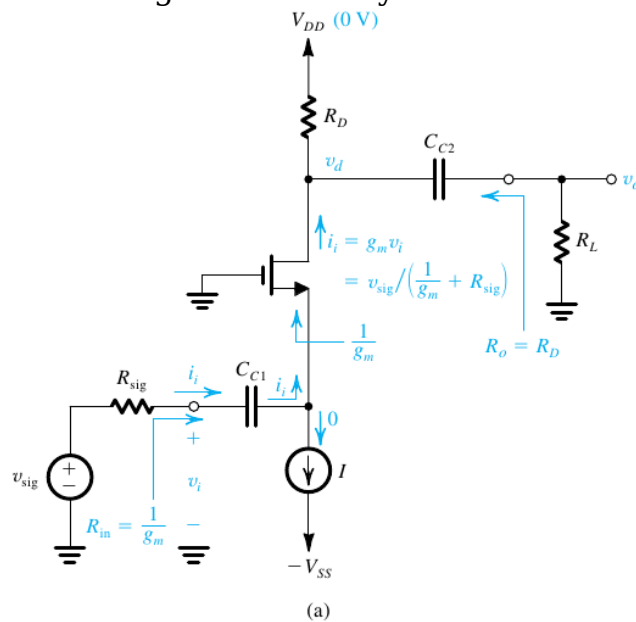


$$\bullet G_v = \frac{v_o}{v_{sig}} = \frac{v_i}{v_{sig}} \frac{v_o}{v_i} = \frac{R_G}{R_G + R_{sig}} \frac{v_o}{v_i} \because v_i = \frac{R_G}{R_G + R_{sig}} v_{sig}$$

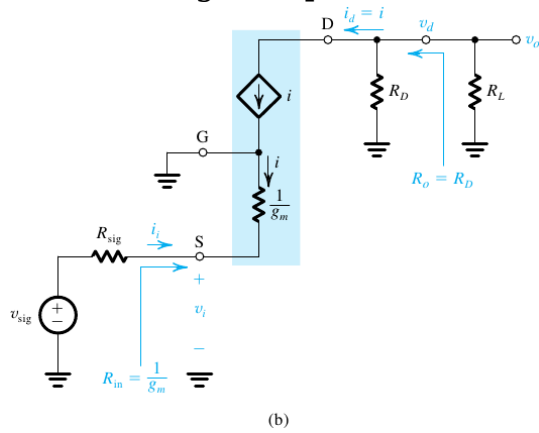
$$\circ G_v = \frac{R_G}{R_G + R_{sig}} A_v = \frac{R_G}{R_G + R_{sig}} \left(\frac{-R_D \parallel R_L}{\frac{1}{g_m} + R_s} \right)$$

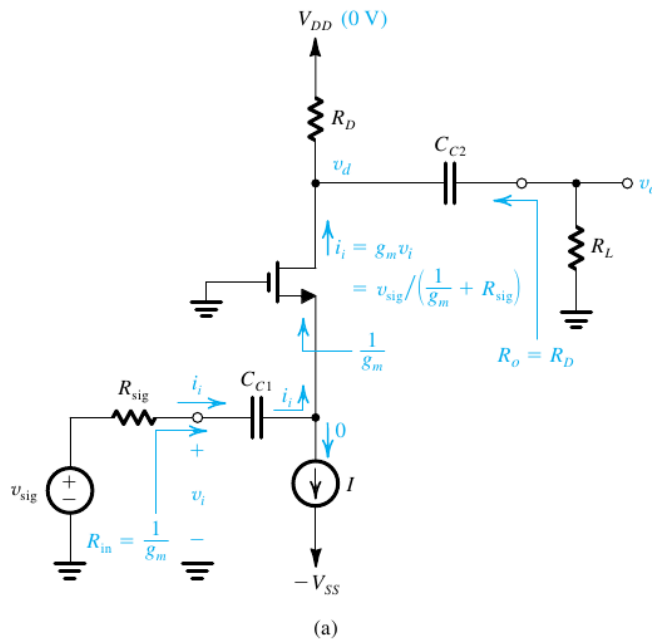
The Common-Gate (CG) Amplifier

- here both the dc and ac voltages at the gate are to be zero,
 - thus the gate is directly connected to the ground

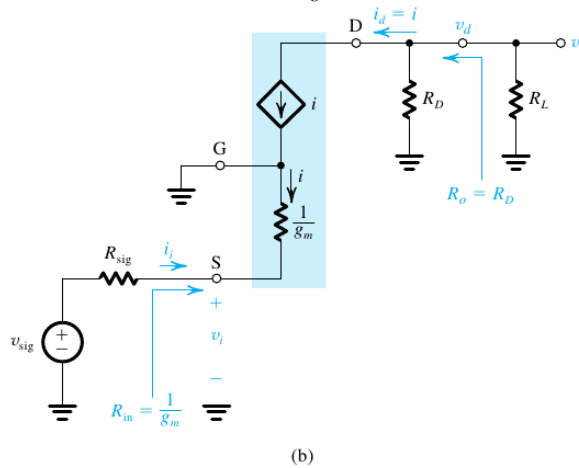


- the small-signal equivalent circuit can be given as





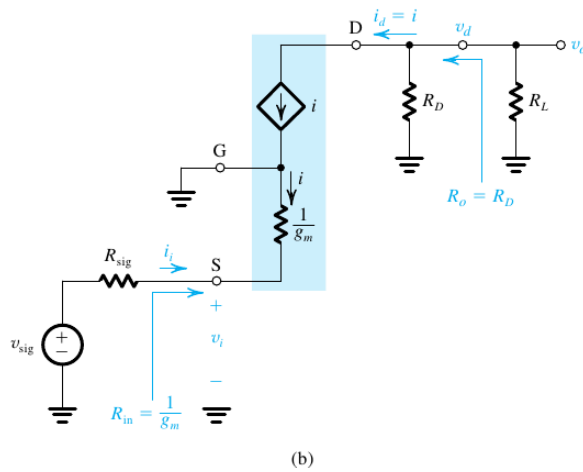
- here $R_{in} = \frac{v_i}{i_i} = \frac{v_i}{-i} = \frac{-v_{gs}}{-i} \because v_i = -v_{gs}$
 - $R_{in} = \frac{-v_{gs}}{-i} = \frac{-v_{gs}}{-g_m v_{gs}} = \frac{1}{g_m}$



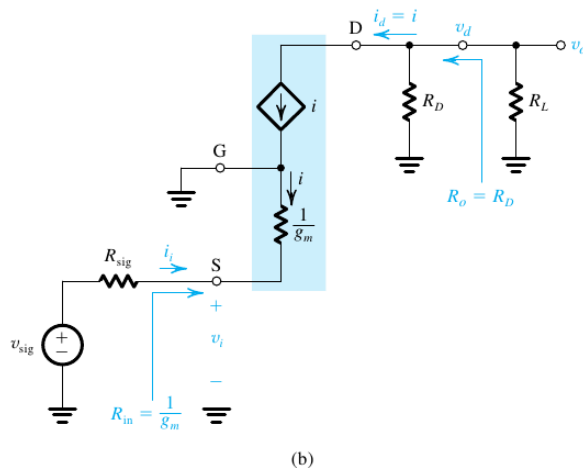
- by voltage divider

$$\circ v_i = \frac{R_{in}}{R_{in} + R_{sig}} v_{sig} = \frac{\frac{1}{g_m}}{\frac{1}{g_m} + R_{sig}} v_{sig}$$

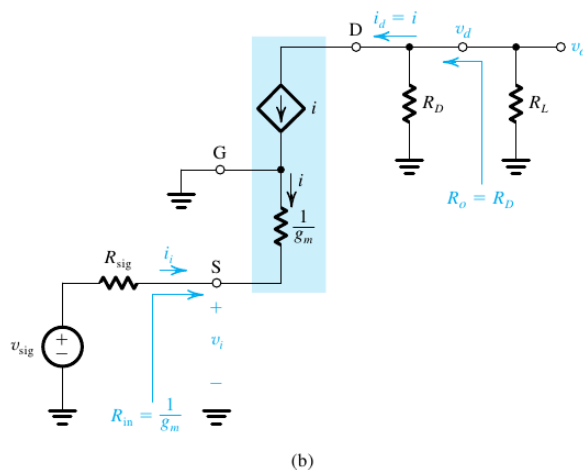
- and $v_i = -v_{gs}$
- to determine R_o , we apply test source at the output and set $v_i = 0$
- $v_i = 0 \Rightarrow v_{gs} = -v_i = 0 \Rightarrow g_m v_{gs} = 0$ or $i = 0$
- $\Rightarrow R_o = R_D$



- $i = g_m v_{gs} = \frac{-v_o}{R_D \parallel R_L}$
 - as $v_{gs} = -v_i$
 - $\Rightarrow i = -g_m v_i = \frac{-v_o}{R_D \parallel R_L}$
 - or $v_o = g_m (R_D \parallel R_L) v_i$
 - $\Rightarrow A_v = \frac{v_o}{v_i} = g_m (R_D \parallel R_L)$
 - and $A_{vo} = A_v|_{R_L \rightarrow \infty} = g_m R_D$

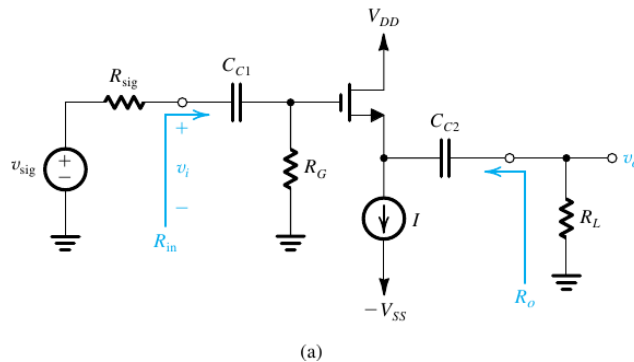


- $A_v = \frac{v_o}{v_i} = g_m (R_D \parallel R_L)$
 - $G_v = \frac{v_o}{v_{sig}} = \frac{v_i}{v_{sig}} \frac{v_o}{v_i} = \frac{v_i}{v_{sig}} A_v$
 - $\Rightarrow G_v = \frac{\frac{1}{g_m}}{\frac{1}{g_m} + R_{sig}} A_v = \frac{\frac{1}{g_m}}{\frac{1}{g_m} + R_{sig}} g_m (R_D \parallel R_L)$
 - or $G_v = \frac{R_D \parallel R_L}{\frac{1}{g_m} + R_{sig}}$

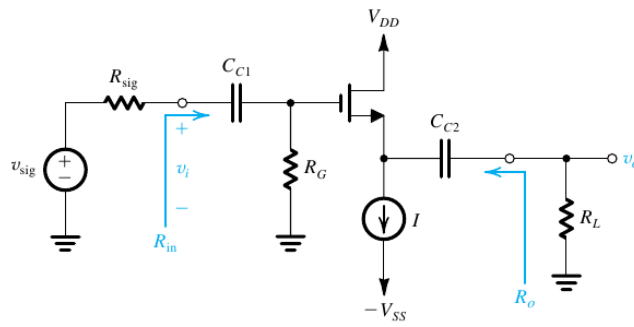


The Source Follower

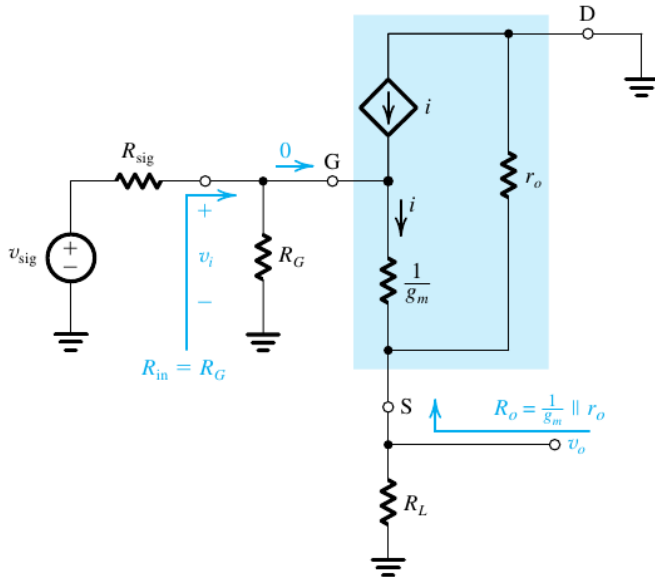
- In this common drain amplifier,
 - As the drain is to function as a signal ground, there is no need for resistor R_D
 - therefore R_D has been eliminated
 - the input signal is applied via C_{C1} to the MOSFET gate
 - the output signal is taken at the MOSFET source and is coupled to the load R_L via C_{C2}



- Small signal equivalent circuit for this amplifier can be obtained
 - by replacing MOSFET with its T-model,
 - by suppressing dc sources
 - and by replacing capacitors by short circuits (as $\frac{1}{j\omega C}$ is very small at the frequency of interest)

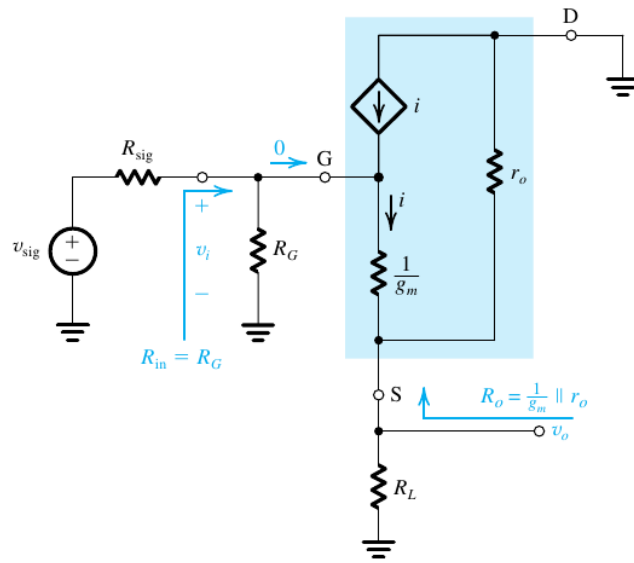


(a)

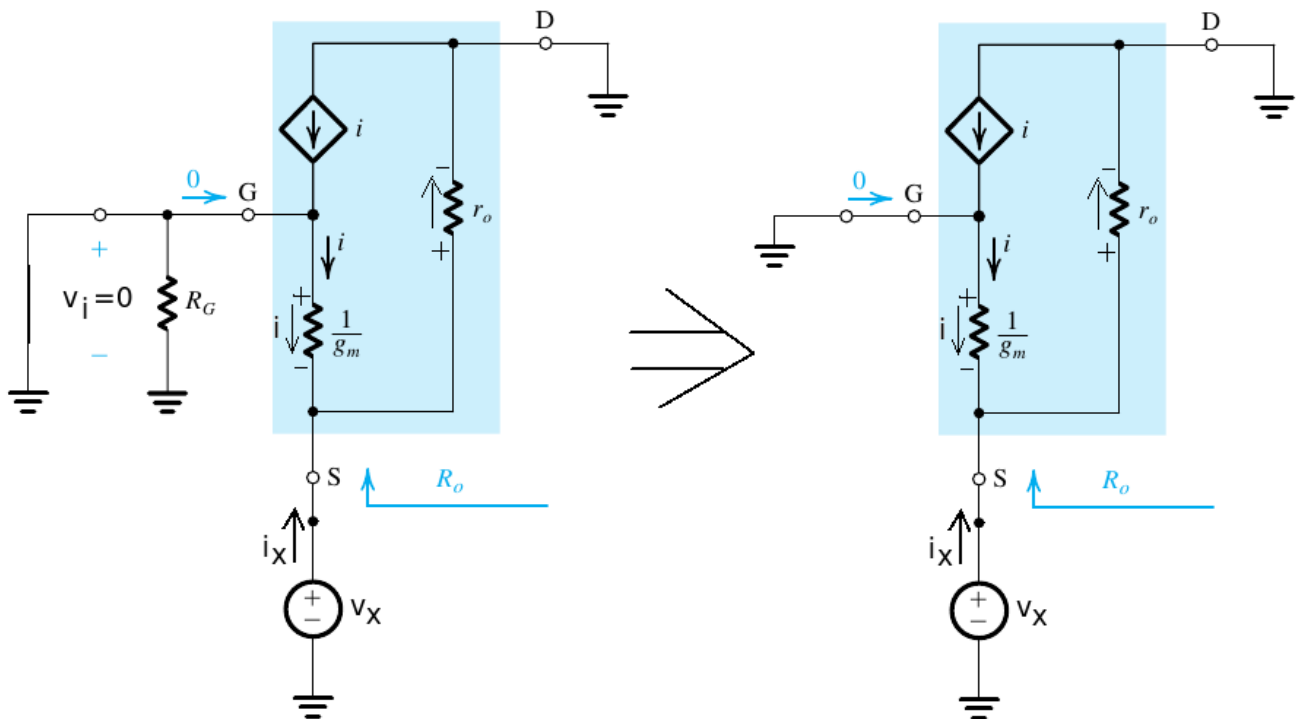


(b)

- here $R_{in} = R_G$
- R_o :
 - to determine R_o , we set $v_i = 0$
 - Remove R_L and apply a test source
 - $v_i = 0 \Rightarrow v_g = 0$



(b)



- KCL at MOSFET Source terminal

$$\circ \Rightarrow i_x + i = \frac{v_x - 0}{r_o} \Rightarrow i_x = \frac{v_x}{r_o} - i$$

- Ohm's Law across resistor $\frac{1}{g_m}$

$$\blacksquare \Rightarrow v_g - v_s = \frac{1}{g_m} i$$

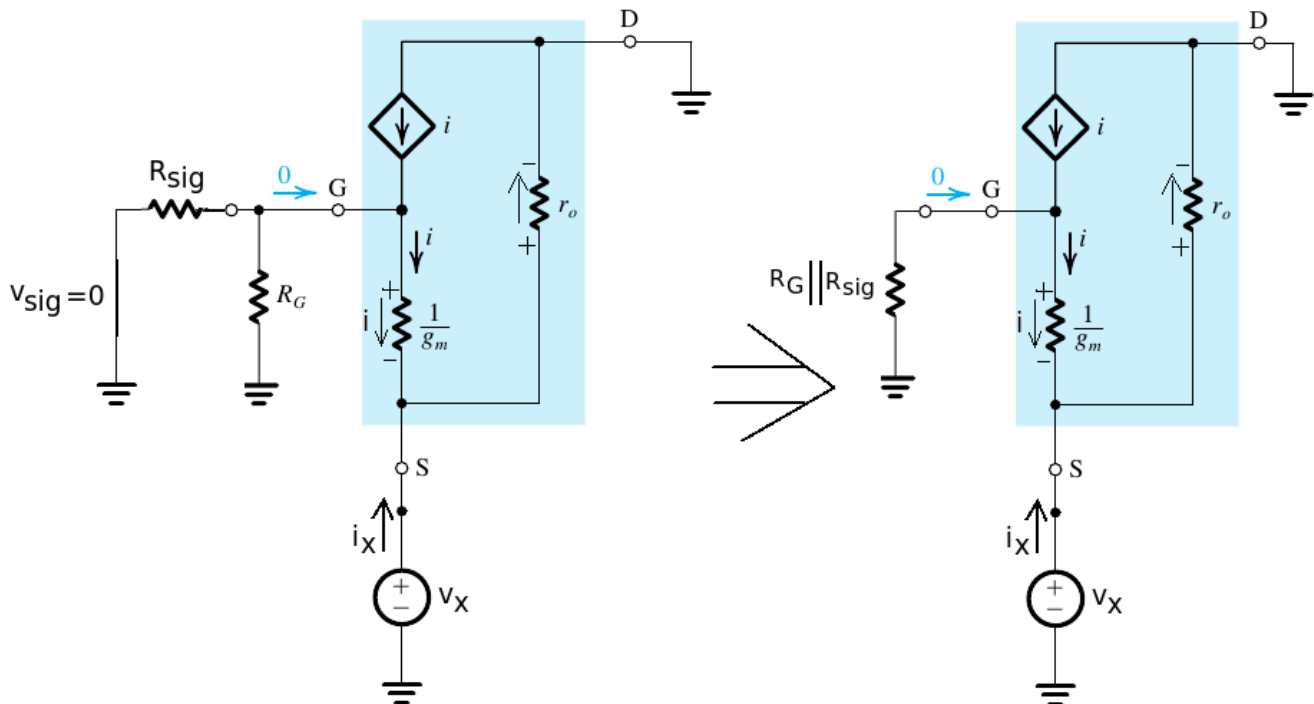
$$\Rightarrow i = g_m (v_g - v_s) = g_m (0 - v_x) = -g_m v_x$$

- eliminating i from above 2 equations

$$\begin{aligned}
 \blacksquare & \Rightarrow i_x = \frac{v_x}{r_o} - i = \frac{v_x}{r_o} - (-g_m v_x) \\
 \blacksquare & i_x = \frac{v_x}{r_o} + g_m v_x = v_x \left(\frac{1}{r_o} + g_m \right) \\
 \blacksquare & \frac{i_x}{v_x} = \frac{1}{r_o} + g_m \\
 \blacksquare & \Rightarrow R_o = \frac{v_x}{i_x} = \frac{1}{\frac{1}{r_o} + g_m} = \frac{1}{\frac{1}{r_o} + \frac{1}{1/g_m}} = r_o \parallel \left(\frac{1}{g_m} \right)
 \end{aligned}$$

R_{out} :

- to determine R_{out} , we set $v_{sig} = 0$, disconnect R_L and apply a test source at the source terminal of MOSFET



- KCL at MOSFET Source terminal

$$\circ \Rightarrow i_x + i = \frac{v_x - 0}{r_o} \Rightarrow i_x = \frac{v_x}{r_o} - i$$

$$\circ \text{ Ohm's Law across resistor } \frac{1}{g_m}$$

$$\blacksquare \Rightarrow v_g - v_s = \frac{1}{g_m} i \Rightarrow i = g_m (v_g - v_s)$$

$$\blacksquare \Rightarrow i = g_m (0 - v_x) = -g_m v_x \because i_g = 0 \Rightarrow v_g = 0$$

- eliminating i from above 2 equations

$$\blacksquare \Rightarrow i_x = \frac{v_x}{r_o} - i = \frac{v_x}{r_o} - (-g_m v_x)$$

$$\blacksquare i_x = \frac{v_x}{r_o} + g_m v_x = v_x \left(\frac{1}{r_o} + g_m \right)$$

$$\blacksquare \frac{i_x}{v_x} = \frac{1}{r_o} + g_m$$

$$\blacksquare \Rightarrow R_{out} = \frac{v_x}{i_x} = \frac{1}{\frac{1}{r_o} + g_m} = \frac{1}{\frac{1}{r_o} + \frac{1}{(1/g_m)}} = r_o \parallel \left(\frac{1}{g_m} \right)$$

A_v :

- By KVL

$$\circ v_i = v_{gs} + v_o = \left(\frac{1}{g_m} \right) i + v_o$$

$$\blacksquare \text{ but } i = \frac{v_o}{R_L \parallel r_o}$$

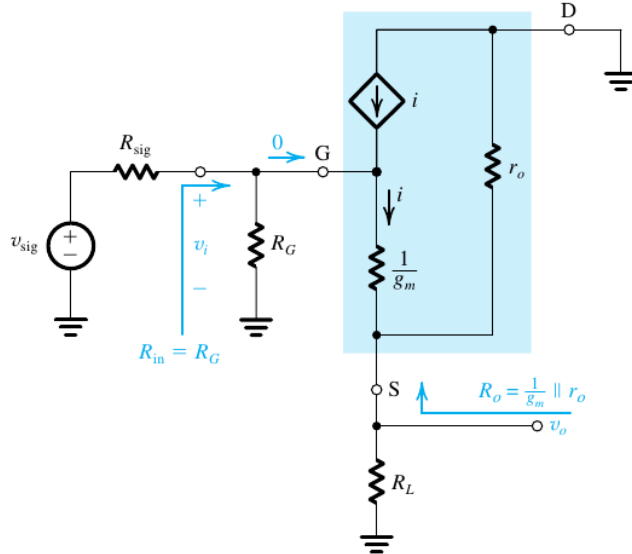
$$\blacksquare \Rightarrow v_i = \frac{1}{g_m} \left(\frac{v_o}{R_L \parallel r_o} \right) + v_o$$

$$\blacksquare v_i = v_o \left(\frac{1}{g_m(R_L \parallel r_o)} + 1 \right)$$

$$\blacksquare v_i = v_o \left(\frac{1 + g_m(R_L \parallel r_o)}{g_m(R_L \parallel r_o)} \right)$$

$$\blacksquare A_v = \frac{v_o}{v_i} = \frac{g_m(R_L \parallel r_o)}{1 + g_m(R_L \parallel r_o)}$$

$$\blacksquare A_{vo} = . A_v|_{R_L \rightarrow \infty} = \frac{g_m r_o}{1 + g_m r_o}$$



(b)

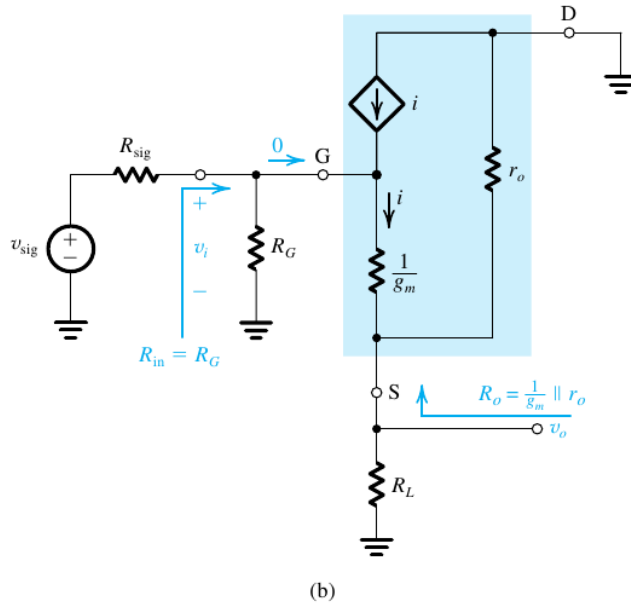
$$\bullet A_v = \frac{v_o}{v_i} = \frac{g_m(R_L \parallel r_o)}{1 + g_m(R_L \parallel r_o)}$$

$$\bullet G_v = \frac{v_o}{v_{sig}} = \frac{v_i}{v_{sig}} \frac{v_o}{v_i} = \frac{v_i}{v_{sig}} A_v = \frac{v_i}{v_{sig}} \frac{g_m(R_L \parallel r_o)}{1 + g_m(R_L \parallel r_o)}$$

$$\circ \text{ by voltage divider } v_i = \frac{R_G}{R_G + R_{sig}} v_{sig}$$

$$\blacksquare \Rightarrow G_v = \frac{R_G}{R_G + R_{sig}} \frac{g_m(R_L \parallel r_o)}{1 + g_m(R_L \parallel r_o)}$$

$$\blacksquare \text{ or } G_v = \frac{R_G}{R_G + R_{sig}} \left(\frac{R_L \parallel r_o}{\frac{1}{g_m} + R_L \parallel r_o} \right)$$



The Amplifier Frequency Response

- Uptill now, we have considered the gain of MOSFET to be constant,
 - and independent of the frequency of the input signal, which is not true
 - A MOSFET has a finite bandwidth (i.e. the gain of MOSFET is depends upon the frequency of the input signal)
 - fig shows the frequency response of a CS amplifier (i.e. gain is plotted versus frequency)

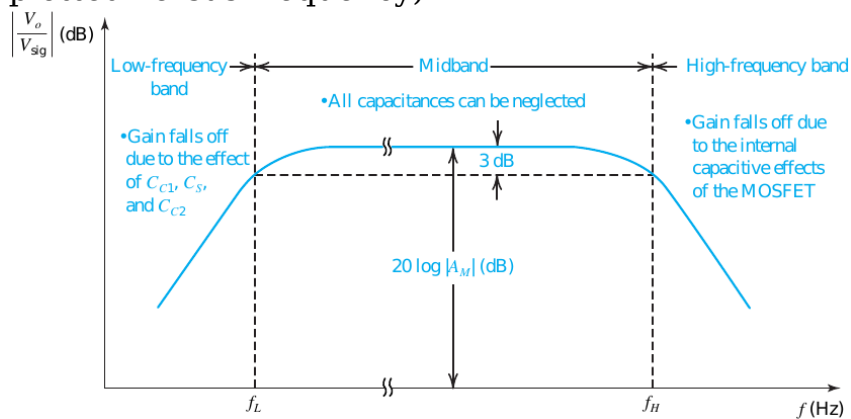
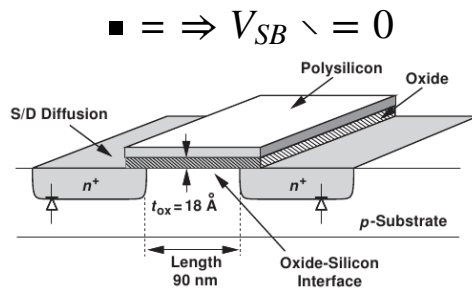


Figure 5.61 A sketch of the frequency response of a CS amplifier delineating the three frequency bands of interest.

- Note that there is a wide frequency range over which the gain remains almost constant
 - this is the useful frequency range (called the midband) of operation for a particular amplifier
- At lower frequencies, the magnitude of amplifier gain falls off
 - because the coupling capacitors, no longer have low impedances
 - i.e. $\frac{1}{j\omega C}$ is large at low frequencies
- At higher frequencies, the gain falls off because of the internal capacitive effects in the MOSFET
 - thus for every MOS amplifier, there is a finite band over which the gain is almost constant.
 - the boundaries of this useful frequency band (called midband), are the two frequencies f_L and f_H
- Note that at f_L and f_H , the gain drops by 3dB below its value at midband
 - thus the amplifier 3dB bandwidth can be defined as
 - $BW = f_H - f_L$
 - as $f_L < f_H \Rightarrow BW \approx f_H$
- A figure of merit for the amplifier is its gain-bandwidth product i.e. $GB = |A_M|BW$
 - where $|A_M|$ is the magnitude of the amplifier gain in the midband
 - this relation $\Rightarrow$ in amplifier design, it should be possible to trade off gain for bandwidth

The Role of the Substrate-The Body Effect

- Thus far, it has been assumed that the source and body terminals are connected $\Rightarrow V_{SB} = 0$
 - with $V_{SB} = 0$, the MOSFET behaves as if it were a three terminal device
 - However, in integrated circuits, the substrate is common to many MOS transistors.
 - In order to maintain the cutoff condition for all the substrate to drain/source junctions,
 - the substrate is connected to the most negative power supply in an NMOS circuit (the most +ve power supply for PMOS devices)



- $V_{SB} \neq 0$

- this non-zero value of V_{SB} affects the i-v characteristics of the MOSFET, by changing its threshold voltage.
- this effect is called the body effect and can be modeled as (for an NMOS)

- $V_t = V_{t0} + \gamma \left[\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f} \right]$
- where V_{t0} is the threshold voltage for $V_{SB} = 0$
- ϕ_f is a physical parameter (with $2\phi_f$ typically 0.6V)

- γ is a fabrication parameter given by

- $\gamma = \frac{\sqrt{2qN_A\epsilon_S}}{C_{ox}}$, where q is the electron charge
- N_A is the doping concentration of the p-type substrate
- ϵ_S is the permittivity of silicon = $11.7\epsilon_o$

- the parameter γ has the dimension of $\sqrt{V}$ and is typically $0.4V^{1/2}$

- for a PMOS transistor

- $V_{tp} = V_{t0} + \gamma \left[\sqrt{2\phi_f + V_{BS}} - \sqrt{2\phi_f} \right]$

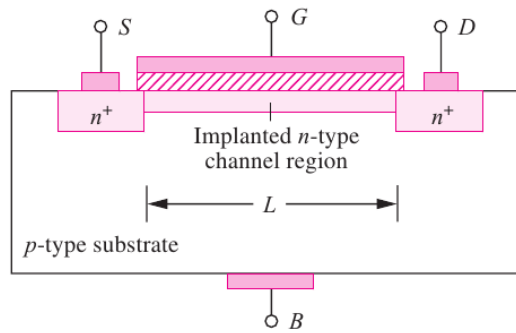
- here γ is -ve and is given as $\gamma = -\frac{\sqrt{2qN_D\epsilon_S}}{C_{ox}}$
- where N_D is the doping concentration of the n-well in which PMOS is formed

- for PMOS transistor

- $2\phi_f$ is typically 0.75V
- γ is typically $-0.5V^{\frac{1}{2}}$

The Depletion-Type MOSFET

- The structure of depletion-type MOSFET is similar to the
 - enhancement-type MOSFET with one important difference
 - The depletion MOSFET has a physically implanted channel



- (b)
- thus an n-channel depletion-type MOSFET has a built-in n-type channel
 - in the device so that the n^+ drain and n^+ source regions are
 - connected through this channel even when $v_{GS} = 0$
- $\Rightarrow$ if a voltage v_{DS} is applied between the drain and source, a current i_D flows for $v_{GS} = 0$
- A negative voltage must be applied to the gate, v_{GS} to deplete the n-type channel region
 - and eliminate the current path between the source and drain
 - (hence the name depletion-type MOSFET)
- thus for depletion-type MOSFET, applying a +ve v_{GS} enhances
 - the channel by attracting more electrons into it
 - and by applying a -ve v_{GS} , electrons can be repelled from
 - the channel and thus the channel will become shallower
 - i.e. the -ve v_{GS} deplete the channel of its charge carriers
 - and this mode of operation (-ve v_{GS}) is called depletion mode
 - As v_{GS} is made more -ve, a value is reached at which
 - the channel is completely depleted of charge carriers $\Rightarrow i_D = 0$
 - and this value of v_{GS} is the threshold voltage of the n-channel depletion type MOSFET
- thus the depletion-type MOSFET can be operated
 - in the enhancement mode by applying a +ve v_{GS}
 - and in the depletion mode by applying a -ve v_{GS}

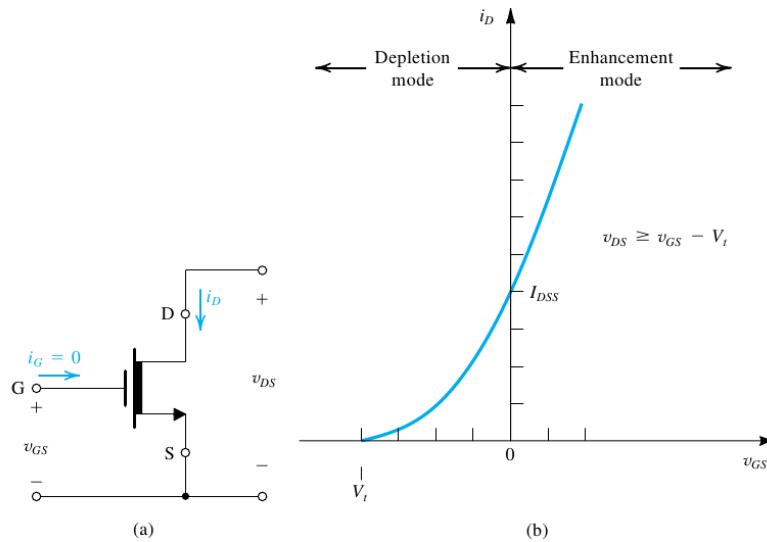
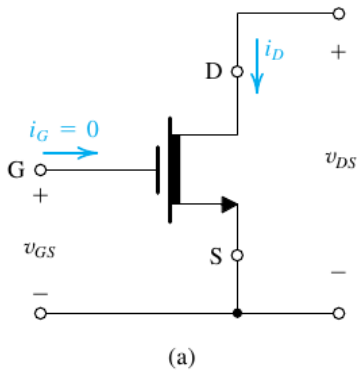


Figure 5.63 The circuit symbol (a) and the i_D - v_{GS} characteristic in saturation; (b) for an n -channel depletion-type MOSFET.



- - Note that the device symbol denotes the existing channel
 - via the shaded area next to the vertical line
- the $i_D - v_{DS}$ characteristics for this device are the same as those for the enhancement type MOSFET
 - except for the negative V_{in}

